

1. Original PoC Scope vs. Expanded Scope

1.1 Original PoC Scope (As Defined)

The initial PoC was designed as a **minimal demonstrable prototype** with:

- **3 User Roles:** Investor, RM, Admin (basic RBAC)
- **Simple Workflows:** Portfolio view, withdrawal request, basic approval
- **No Email/Notification System**
- **No Document Management**
- **No Lifecycle State Management**
- **No Contract Management**
- **Timeline:** 4-6 weeks ()
- **Effort Estimate:** 320-400 development hours

1.2 Expanded Scope (Based on New Requirements)

The expanded scope now includes:

Feature Area	Original PoC	New Requirements
User Lifecycle Management	Simple user creation	6-stage lead-to-client pipeline with automated state transitions
DocAdmin Role & Workflows	Not included	Complete DocAdmin portal with 5 dedicated tabs, RM assignment, KYC verification, contract management
RM Dashboard Views	1 simple view	6 dedicated filtered views with status tracking and follow-up management
Time-Based KYC Rules	Not included	7-day automated timeline with reminder emails, account state changes, archival logic
Purchase Request Flow	Basic transaction	Multi-stage approval workflow: Client → RM → DocAdmin → Contract Upload → Pre-signed Contract → Finalization

Contract Management System	Not included	Contract creation, upload, validity tracking, signature collection, finalization, storage
Email Notification System	Not included	10+ automated email triggers across entire user journey
Product Page & Cart UI	Static product list	Interactive slider, real-time ROI calculator, FAQ system, dynamic calculations
Registration Without RM	Not considered	Conditional access logic, banner notifications, restricted functionality
Data Archival & Cleanup	Not included	Automated archival of expired KYC accounts to separate tables

2. Detailed Impact Analysis by Module

2.1 New Enquiry (Lead) Lifecycle Management

Original PoC: Direct user creation with immediate portfolio access.

New Requirements:

- 6-stage lifecycle: New Enquiry → RM Assigned → KYC Pending → KYC Approved → Contract Pending → Completed
- State machine logic with validation rules at each transition
- Automated state progression based on document submission and approval events
- Historical state tracking and audit trail

Technical Implications:

- Database schema redesign to accommodate lead lifecycle states
- State transition engine with business rule validation
- Workflow orchestration layer

- Migration strategy for existing mock data

Additional Effort: 80-100 hours

2.2 DocAdmin Role & Dashboard (5 Dedicated Tabs)

Original PoC: Admin had basic CRUD operations only.

New Requirements:

Tab 1: New Enquiries

- Display all leads with form data
- RM assignment interface with dropdown/search
- Bulk assignment capability
- Lead status filtering and sorting

Tab 2: KYC Pending

- Document preview interface
- Verification checklist
- Approve/Reject workflow with comments
- Document versioning

Tab 3: Product Requested

- Client-product mapping display
- Investment amount details
- Status tracking per product request

Tab 4: Contract Pending

- Contract upload interface (PDF)
- Product and terms selection UI
- Validity period configuration
- Pre-signed contract upload for RM and client signatures

Tab 5: Completed

- Finalized contracts repository
- Search and filter by client/date/product
- Transaction creation trigger
- Portfolio update automation

Technical Implications:

- 5 distinct UI components with unique data models
- Document storage and retrieval system

- PDF preview and annotation capabilities
- Complex permission matrix (DocAdmin-specific actions)
- Integration with transaction and portfolio engines

Additional Effort: 120-140 hours

2.3 Enhanced RM Dashboard (6 Dedicated Views)

Original PoC: Single view of assigned clients with basic portfolio data.

New Requirements:

View 1: Leads (Not Registered)

- Contact information display
- Follow-up notes system (CRUD)
- Status management (New / Contacted / Interested / Lost)
- Activity timeline

View 2: Registered Clients (No KYC)

- Email verification status
- Push KYC reminder button
- Assignment confirmation

View 3: KYC Pending Clients

- KYC Pending Clients list
- Real-time KYC status from DocAdmin
- Cannot approve (read-only)

View 4: Active Clients

- Fully verified client list
- Portfolio summary cards

View 5: Purchase Requests

- Pending purchase approvals
- Client details, product, amount
- Approve/Reject with notes

View 6: Withdrawal Requests

- Pending withdrawal approvals

- Transaction history context
- Approve/Reject workflow

Technical Implications:

- 6 independent data queries with optimized indexing
- Real-time status synchronization across modules
- Notes and activity tracking system
- Responsive UI for each view with filtering/sorting
- Permission checks for each action

Additional Effort: 100-120 hours

2.4 Time-Based KYC Rules & Automation

Original PoC: Manual KYC flow with no time constraints.

New Requirements:

- Day 0: Email Verified
 - User can log in
 - RM is assigned (by DocAdmin)
- Day 3
 - First KYC reminder email is sent
 - In-app alert is triggered
 - RM can contact the client
- Day 6
 - Final reminder is sent, along with a deactivation warning
- Day 7: Account State
 - Client cannot transact
 - Account is marked as KYC_EXPIRED

Technical Implications:

- Background job scheduler (cron/queue-based)
- Email service integration (SendGrid/AWS SES)
- Database table for archived accounts
- State change automation
- Notification service with template management

Additional Effort: 60-80 hours

2.5 Multi-Stage Purchase Request Flow

Original PoC: Simple purchase creation with admin approval.

New Requirements:

Step-by-Step Flow:

1. **Client Submission:** Product selection + amount
2. **RM Review:** Approve/Reject
3. **Pre-Signed Contract Upload:** DocAdmin uploads contracts pre-signed by RM and client after RM approval
4. **Finalization:** DocAdmin finalizes → Transaction created, portfolio updated
5. **Contract Period Start:** Same day as finalization

Technical Implications:

- Multi-stage approval engine with conditional routing
- Transaction state management (Pending → RM Approved → Contract Uploaded → Pre-Signed → Finalized)
- Contract validity period calculation
- Portfolio update automation on finalization
- Rollback mechanisms for rejected requests
- Email notifications at each stage

Additional Effort: 80-100 hours

2.6 Contract Management System

Original PoC: No contract management.

New Requirements:

- Pre-signed contract storage (separate field/version)
- Product and terms association
- Validity period tracking
- Contract repository with search/filter
- Contract expiry notifications (60 days before end date)

Technical Implications:

- Document management system (DMS) integration
- Secure storage with encryption
- Metadata management (product, terms, validity)

Additional Effort: 70-90 hours

2.7 Comprehensive Email Notification System

Original PoC: No email system.

New Requirements - 19 Email Triggers:

A. Client (End User) Emails

- **Registration & Verification:**
 - Email verification
 - Welcome email and notification for KYC
- **KYC & Account Status:**
 - KYC reminder – Day 3
 - KYC final reminder + deactivation warning – Day 6
 - KYC final + deactivation email (on account expiry)
 - KYC rejection notification
- **Transactions & Contracts:**
 - Purchase request submitted
 - Purchase approved / rejected
 - Contract shared
- **Post-Investment & Withdrawal:**
 - Payout date reminder
 - Contract renewal reminder (60 days before expiry)
 - Withdrawal approved / escalated to Admin

B. Relationship Manager (RM) Emails

- New lead assigned
- Client registered but KYC pending
- Purchase request received
- Contract signed notification
- Withdrawal request submitted by client

C. Document Admin (DocAdmin) Emails

- KYC documents submitted by client

D. Admin Emails

- Withdrawal request escalated by RM
- 1.

Technical Implications:

- Email service provider integration (SendGrid/AWS SES)
- Template management system
- Dynamic content rendering
- Email queue management
- Delivery tracking and retry logic
- Email logs and audit trail

Additional Effort: 50-70 hours

2.8 Product Page & Cart UI Enhancements

Original PoC: Basic product list with details.

New Requirements:

- Enhanced product listing with key details highlighted
- Dedicated FAQ section per product
- Interactive investment amount slider on cart page
- Dynamic return projections with visual charts
- Responsive and accessible UI

Technical Implications:

- Advanced UI component library (charts, sliders)
- Real-time calculation engine
- FAQ management
- Responsive design for mobile/tablet
- Performance optimization for calculations

Additional Effort: 40-60 hours

2.9 Registration Without RM (Conditional Access)

Original PoC: Not considered.

New Requirements:

- Client can log in after email verification
- Can see product
- Cannot request products or submit purchases
- Banner notification: "RM not assigned / KYC pending"
- Conditional UI rendering based on account state
- Restricted navigation

Technical Implications:

- Frontend permission middleware
- Conditional rendering logic
- Banner notification component
- State-based route guards

Additional Effort: 20-30 hours

2.10 Data Archival & Cleanup Logic

Original PoC: No data archival.

New Requirements:

- After Day 7 (KYC expired), move user data to separate archive table
- Do not send further emails to archived users
- Reactivation workflow if user returns

Technical Implications:

- Separate database table for archived accounts
- Automated migration job
- Data integrity maintenance
- Reactivation logic and testing

Additional Effort: 30-40 hours

3. Payout & Withdrawal Request Clarifications

Current Ambiguity: The requirements mention "Payout dates (If any) 15th of next month" and withdrawal requests, but the exact payout logic and withdrawal workflow details are not fully specified.

Required Clarifications:

1. **Payout Schedule:**

- Is there a fixed monthly payout for all active contracts?
- Is payout interest-only or principal + interest?
- When does the payout cycle start (contract start date or fixed calendar date)?

2. **Withdrawal Request Flow:**

- Can clients withdraw before contract maturity?
- Are there penalties for early withdrawal?
- What is the approval hierarchy (RM → DocAdmin → Finance)?
- What is the expected settlement timeline?

3. **Maturity Handling:**

- What happens when a contract reaches maturity?
- Auto-renewal or manual renewal decision?
- Notification workflow 60 days before maturity (already specified)

Recommendation: Schedule a detailed discussion on payout and withdrawal logic to avoid scope gaps during development.

Potential Additional Effort (if complex): 40-60 hours

4. Technical Infrastructure Expansion

4.1 Database Schema Changes

- Lead lifecycle states table
- DocAdmin actions audit log
- Contract metadata table
- Email log table
- Archived accounts table
- Product FAQ table
- RM notes and follow-ups table

Effort: 30-40 hours for schema design, migration scripts, and indexing.

4.2 API Development

- **25+ new API endpoints** for:
 - DocAdmin workflows
 - RM dashboard views
 - Lead lifecycle management
 - Contract operations
 - Email trigger endpoints
 - KYC automation

Effort: 80-100 hours for API development, validation, and documentation.

4.3 Background Job System

- **Scheduled jobs** for:
 - KYC reminder emails (Day 3, 6, 7)
 - Account archival (Day 7)
 - Payout reminders (15th of month)
 - Contract renewal reminders (60 days before expiry)

Effort: 40-50 hours for job scheduler setup, queue management, and monitoring.

4.4 Document Storage & Management

- **AWS S3 or equivalent** for:

- KYC documents
- Contract PDFs
- Pre-signed contracts
- Audit documents

Effort: 30-40 hours for integration, security setup, and backup strategy.

4.5 Email Service Integration

- **SendGrid/AWS SES** setup
- **Template creation** (10+ templates)
- **Delivery tracking and analytics**

Effort: 20-30 hours for integration and testing.

**FROM HERE THERE IS OTHER NON TECHNICAL AND NON FUNCTIONAL
THING I DO NOT THINK THIS IS SOMETHING TO SHARE SO WE CAN REMOVE
THIS**

5. Quality Assurance & Testing Expansion

5.1 Additional Test Scenarios

- **State transition testing** for lead lifecycle (54 test cases)
- **Time-based automation testing** (KYC reminders, archival)
- **Multi-stage purchase flow testing** (25 test cases)
- **Email delivery and template testing** (10+ scenarios)
- **DocAdmin permission matrix testing** (30+ scenarios)
- **RM dashboard data accuracy testing** (6 views × 5 scenarios = 30 test cases)

Effort: 80-100 hours for comprehensive QA across all new modules.

5.2 Security & Compliance Testing

- **Role-based access control validation** across all new roles
- **Document storage security audit**
- **Email system security** (prevent spam, ensure encryption)
- **Data archival compliance** (GDPR/data retention)

Effort: 40-50 hours for security testing and compliance review.

6. Revised Timeline & Milestones

Phase	Deliverables	Duration	Dependencies
Phase 1: Requirements Finalization	Payout/withdrawal logic clarification, wireframe approval	1 week	Client input required
Phase 2: Database & Architecture	Schema redesign, API architecture, infrastructure setup	2 weeks	Phase 1 complete
Phase 3: Core Development - Backend	Lead lifecycle, DocAdmin APIs, RM APIs, Contract management, Email system	5 weeks	Phase 2 complete
Phase 4: Core Development - Frontend	DocAdmin portal (5 tabs), RM dashboard (6 views), Enhanced product/cart UI	4 weeks	Phase 3 (partial overlap)
Phase 5: Automation & Jobs	KYC reminders, archival jobs, payout reminders, contract renewal reminders	2 weeks	Phase 3 complete
Phase 6: Integration & Testing	End-to-end workflow testing, email delivery testing, security audit	3 weeks	Phase 4 complete
Phase 7: UAT & Deployment	User acceptance testing, bug fixes, production deployment	2 weeks	Phase 6 complete
Phase 8: Documentation & Training	User manuals, admin guides, RM training, client onboarding docs	1 week	Phase 7 complete

Total Timeline: 20 weeks (approximately 5 months)

Original PoC Timeline: 4-6 weeks

Timeline Expansion: 3.3x - 5x

7. Revised Effort & Resource Estimates

7.1 Development Effort Breakdown

Module/Feature	Original PoC Effort (Hours)	Expanded Scope Effort (Hours)	Delta
User Management & RBAC	40	80	+40
Lead Lifecycle Management	0 (not included)	100	+100
DocAdmin Portal (5 Tabs)	0 (not included)	140	+140
RM Dashboard (6 Views)	40 (1 view)	120	+80
Time-Based KYC Automation	0 (not included)	80	+80
Multi-Stage Purchase Flow	40 (simple flow)	100	+60
Contract Management System	0 (not included)	90	+90
Email Notification System	0 (not included)	70	+70
Product Page & Cart UI Enhancements	30 (basic list)	60	+30

Registration Without RM Logic	0 (not included)	30	+30
Data Archival & Cleanup	0 (not included)	40	+40
Payout & Withdrawal Logic (TBD)	0	60 (estimated)	+60
Database Schema & Migrations	30	70	+40
API Development	80	180	+100
Background Jobs System	0 (not included)	50	+50
Document Storage Integration	0 (not included)	40	+40
Email Service Integration	0 (not included)	30	+30
QA & Testing	60	160	+100
Security & Compliance Testing	20	70	+50
Deployment & DevOps	30	60	+30
Documentation & Training	20	60	+40

Total	390 hours	1,580 hours	+1,190 hours
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Effort Expansion: 4x

7.2 Team Composition (Revised)

Role	Allocation	Duration
Project Lead	50%	20 weeks
Backend Developer (Senior)	100%	18 weeks
Backend Developer (Mid-Level)	100%	16 weeks
Frontend Developer (Senior)	100%	18 weeks
Frontend Developer (Mid-Level)	100%	14 weeks
UI/UX Designer	50%	12 weeks
QA Engineer	100%	12 weeks
DevOps Engineer	50%	10 weeks
Technical Writer	25%	4 weeks

Original PoC Team: 3-4 developers, 1 designer (part-time)

Revised Team: 8-9 resources with increased allocation



8. Commercial Impact Analysis

8.1 Original PoC Investment Estimate

- **Development Hours:** 390 hours
- **Team Size:** 3-4 resources
- **Timeline:** 4-6 weeks
- **Estimated Cost Range:** ₹X - ₹Y (baseline)

8.2 Revised Investment Estimate (Expanded Scope)

- **Development Hours:** 1,580 hours
- **Team Size:** 8-9 resources
- **Timeline:** 20 weeks (5 months)
- **Estimated Cost Range:** ₹4X - ₹5Y (4-5x baseline)

8.3 Cost Breakdown by Category

Category	Original PoC (%)	Expanded Scope (%)	Notes
Backend Development	35%	40%	Complex workflow engine, state management
Frontend Development	30%	30%	Multiple dashboards, enhanced UI
Infrastructure & DevOps	10%	12%	Document storage, job scheduler, email service
QA & Testing	15%	12%	Comprehensive test coverage
Design	5%	3%	More screens but streamlined design system

Project Management & Documentation	5%	3%	Longer timeline but proportional allocation
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8.4 Additional Operational Costs

- **Email Service:** ₹X/month (SendGrid/AWS SES)
- **Document Storage:** ₹Y/month (AWS S3 with versioning)
- **Background Job Queue:** ₹Z/month (AWS SQS or equivalent)
- **Monitoring & Logging:** ₹A/month (CloudWatch or equivalent)

Monthly Operational Cost Increase: ₹(X+Y+Z+A)

9. Risk Assessment & Mitigation

9.1 Key Risks

Risk	Impact	Mitigation Strategy
Scope Creep	Further delays and cost overruns	Formal change control process, weekly scope reviews
Payout/Withdrawal Logic Ambiguity	Development rework	Immediate clarification meeting, detailed workflow documentation
Email Deliverability Issues	Poor user experience, missed notifications	Early email service integration, SPF/DKIM setup, delivery monitoring
Performance with Large Data Sets	Slow dashboard loading	Database indexing, caching strategy, pagination
Contract Management Complexity	Development delays	Phased rollout: manual upload first, then automation

Timeline Pressure

Quality compromise

Realistic timeline communication,
prioritized feature rollout

10. Phased Delivery Recommendation

Given the significant scope expansion, we recommend a **phased delivery approach** to balance time-to-market with comprehensive feature delivery:

Phase 1: Core Platform (Weeks 1-12)

- Lead lifecycle management (6 stages)
- DocAdmin portal (Tabs 1, 2, 5)
- RM dashboard (Views 1, 3, 4)
- Basic purchase flow (without pre-signed contracts)
- Essential email notifications (registration, KYC, purchase)

Deliverable: Functional platform for investor meetings with core workflows demonstrated.

Phase 2: Enhanced Workflows (Weeks 13-16)

- DocAdmin Tabs 3, 4 (Product Requested, Contract Pending)
- RM dashboard Views 2, 5, 6
- Contract management with pre-signed upload
- Time-based KYC automation
- Complete email notification system

Deliverable: Production-ready platform with all lifecycle management features.

Phase 3: User Experience & Optimization (Weeks 17-20)

- Product page & cart UI enhancements (slider, ROI calculator, FAQ)
- Payout and withdrawal workflows (after clarification)
- Data archival and cleanup automation
- Performance optimization and security hardening
- Documentation and training materials

Deliverable: Fully optimized platform ready for client onboarding and scaling.

11. Client Decision Points

To proceed with the expanded scope, we require decisions on the following:

11.1 Immediate Decisions Required

1. **Payout & Withdrawal Logic:** Schedule detailed discussion to finalize workflows and business rules.
2. **Pre-Signed Contract Workflow:** Confirm process for collecting RM and client signatures (manual upload or digital signature integration?).
3. **Data Archival Policy:** Confirm retention period for archived KYC-expired accounts (30 days? 90 days? 1 year?).
4. **Budget Approval:** Approve revised commercial estimate or discuss phased delivery options.
5. **Timeline Acceptance:** Confirm acceptance of 20-week timeline or prioritize features for faster Phase 1 delivery.

11.2 Optional Enhancements (Future Consideration)

- Digital signature integration for contracts (reduces manual pre-signing)
 - Mobile app for RM and client (React Native)
 - Advanced analytics dashboard with AI-driven insights
 - Multi-currency support
 - WhatsApp notifications in addition to email
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12. Conclusion

The expanded scope represents a **4x increase in development effort** and a **3.3x-5x increase in timeline** compared to the original PoC plan. This is driven by:

1. Introduction of **7 major new modules** (DocAdmin portal, lead lifecycle, contract management, email system, etc.)
2. Expansion of RM dashboard from **1 view to 6 dedicated views**
3. Addition of **time-based automation** for KYC and notifications
4. Implementation of **multi-stage approval workflows** for purchases
5. **Enhanced UI/UX** with interactive calculators and product enhancements

The revised platform will deliver a **production-grade, scalable, and investor-ready** wealth management system that fully supports the operational needs of Investors, RMs, and DocAdmins across the entire client lifecycle.

We recommend proceeding with a **phased delivery approach** to demonstrate value early while building toward the complete vision. This approach allows for:

- Faster time-to-investor-demo (Phase 1 in 12 weeks)
 - Incremental feedback incorporation
 - Budget flexibility across phases
 - Reduced delivery risk
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13. Next Steps

1. **Schedule Requirements Clarification Meeting** (Week 1)
 - Finalize payout/withdrawal workflows
 - Confirm contract signature collection process
 - Clarify data archival policies
 2. **Review & Approve Revised Timeline** (Week 1)
 - Confirm 20-week timeline or discuss accelerated Phase 1 delivery
 - Align on phase-wise milestones
 3. **Approve Revised Commercial Estimate** (Week 1-2)
 - Review detailed cost breakdown
 - Finalize payment milestones aligned with delivery phases
 4. **Kickoff Phase 1 Development** (Week 2)
 - Begin database schema design and infrastructure setup
 - Start frontend wireframe approvals for DocAdmin and RM dashboards
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